ECE 382 N **Vijay Garg** **Fall 2026**

Term Paper

A term paper is required for EE 382N. The paper will be due on November 19. It should be approximately 6 pages excluding references. The source code developed for the project should be submitted as a soft copy. Many projects require you to implement multiple algorithms. The goal is to implement and evaluate the performance of these algorithms.

A project should be done by a team of two students. The list of projects given below is just suggestive; you can work on any other topic related to distributed systems. You can sign up for a project by adding your name after the project title.

Distributed Trigger Counting Algorithms:

The goal of this project is to improve or simplify algorithms for distributed trigger counting. The starting point for this project is going to be the following algorithms: LayeredRand, CoinRand and RingRand. The open problems are: can these algorithms be derandomized? Can the efficiency/analysis be improved by using power of two choices in randomized load balancing. There is also no experimental study of these algorithms. click here

Vector Clock Algorithms – Abhay & Ishan

The goal of this project is to improve or simplify vector clock algorithms for scalable distributed/parallel systems. There are many possible directions.

- This paper gives an algorithm for chain construction based on vertex cover of all events in the computation. Can you generalize the algorithm for online/offline scenario by using strings? (See the paper that appeared in .)
- This paper gives an online algorithm for chain construction. Can you generalize the algorithm by weakening requirements on event arrival?
- This paper gives the concept of a string realizer of a poset. Can you design hierarchical vector clocks that reduce the vector clock overhead for systems that are hierarchical in nature, i.e., most communication occurs within the local groups.
- This paper gives an incremental algorithm for chain decomposition. Can you improve the implementation or the algorithm for the special case when most events cover only two or less events.
- This paper introduces the notion of rectangle realizers. Can you design an offline or an online algorithm to develop timestamping based on rectangles.
- This paper gives an algorithm to timestamp events in synchronous computations. Can the algorithm be improved?

Global Predicate Detection Algorithms -

Design and implement a distributed algorithm for detecting global predicates expressed in BTL (Basis Temporal Logic) as specified in the DISC'07 paper by Ogale and Garg. pdf

Fusion based Fault-Tolerance

Develop a P2P (Chord-like system) that uses fusion to provide fault-tolerance. For fusion, click here For Chord, click here

Error Correcting Codes for Fault-tolerance

Compare performance of various coding algorithms for maintaining fused tables as backups. For fusion, click here

Weighted Byzantine Agreement

Implement and evaluate weighted Byzantine Agreement algorithms. For some examples of weighted Byzantine Algorithm click here Can you design a weighted Byzantine agreement algorithm based on Gradecast? Gradecast paper is available here

Accurate Byzantine Agreement with Feedback -

Design and implement algorithms for Accurate Byzantine agreement. Compare various weight-update strategies. click here

Byzantine Vector Consensus –

Do a comparative evaluation of approximate Byzantine Vector Consensus algorithms by Vaidya and Garg [PODC 2013] and by Hammurabi and Herlihy [STOC 2013]. Can you improve their results? a paper that contains both algorithms

Distributed Algorithms for Lattice Enumeration

Implement distributed algorithms for enumerating lattice of consistent cuts in lexical manner. The following papers describe a centralized algorithm and a parallel algorithm for lexical enumeration. click here and here

Distributed Algorithm for Relational Predicates -

Implement offline, and online (central and distributed) algorithms for detection of relational predicates using max-flow techniques. The algorithm is presented here

Distributed Algorithm for Slicing

Implement offline, and online (central and distributed) slicing algorithms for detection of regular predicates. The implementation should be able to monitor overall memory and CPU usage, as well as number of messages exchanged by the algorithm. The algorithm is presented in SRDS 2013 paper

Controlled Reexecution -

Design a system that implements controlled reexecution as outlined in Chapter 15. Alternatively, extend the algorithms for different classes of predicates. A paper that describes controlled execution is here

Predicate Detection in the Locking Poset Model

A locking poset models the locking constraints in addition to the happened-before. Investigate predicate detection in this model. The model is described here

Democratic Election in Distributed Systems

Leader election in distributed systems may sometimes require us to take into consideration the preferences of the processes. This leads to the problem of conducting democratic elections in distributed systems in the form of social choice and social welfare functions with three or more candidates. An algorithm to conduct democratic election is presented in here

Distributed Algorithms for Stable Matching Problems

In the standard stable matching problem, there are n men and n women each with their totally ordered preference list. The goal is to find a matching between men and women such that there is no instability, i.e., there is no pair of a woman and a man such that they are not married to each other but prefer each other over their partners. The constrained stable matching problem requires the matching to satisfy additional constraints. Implement distributed algorithms for the constrained stable matching problem. One such proposal is presented here

Distributed Algorithm for Improved Path to the Stable Matching Problems -

In the standard stable matching problem, there are n men and n women each with their totally ordered preference list. Instead of starting from the top preferences for men, we start with an arbitrary assignment of men to women. Our goal is to find a matching between men and women that is close to the initial assignment. Roth and Vande Vate give an algorithm of length $O(n^3)$. We present an algorithm that achieves stability in a sequence of matchings of length $O(n^2)$. One such proposal is presented in here Give a distributed version of this algorithm.

Distributed Algorithms for Lattice Agreement

We consider the lattice agreement problem (a weaker version of the standard agreement problem). We give an $O(\log n)$ algorithm for lattice agreement. Give a lower bound on the problem (or, improve the upper bound). A version is available here .

Distributed Algorithms for Lattice Agreement in Byzantine Systems

We consider the Byzantine lattice agreement problem. We give an $O(\log n)$ algorithm for lattice agreement in synchronous systems and asynchronous systems. The version in the synchronous system uses $4 \log f +3$ messages and tolerates $n/3$ failures. Can it reduced to $\log f$ messages? Synchronous Systems The asynchronous version requires less than $n/5$ faults. Can it be improved to handle $n/3$ faults? Asynchronous Systems

Distributed Algorithms for Weighted Bipartite Matching

The hungarian algorithm is a well-known algorithm to solve the weighted bipartite matching (or equivalently the assignment) problem. Design distributed algorithms to solve the weighted bipartite matching problem. A possible implementation based on lattice-linear predicate detection is here .

Self-stabilizing Algorithms

Design and implement self-stabilizing algorithms such as an algorithm to maintain a spanning tree in a distributed system. One such algorithm is here

Distributed Algorithms for the Lattice-Linear Predicate Detection

The following paper gives a parallel algorithm to solve the shortest path problem. Give a distributed version of the algorithm here .

Distributed Algorithms for the Popular Matching Problem

The following paper gives a parallel algorithm to solve the popular matching problem. Give a distributed version of the algorithm here .

Distributed Algorithms for the Housing Allocation Problem

The following paper gives a distributed algorithm to solve the housing allocation problem. Give an efficient implementation of the algorithm presented here .